Worksheet 22

November 23, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Double-angle values

Suppose $\sin\theta = \frac{5}{13}$ and $\frac{\pi}{2} < \theta < \pi$.

(a) Find the exact value of $\cos\theta$, using the quadrant to choose its sign.

(b) Find $\sin(2\theta)$ and $\cos(2\theta)$ exactly.

Worksheet 22

November 23, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. Half-angle signs

Suppose $\cos\theta = -\frac{3}{5}$ and $\pi < \theta < \frac{3\pi}{2}$.

(a) Identify the quadrant of $\frac{\theta}{2}$.

(b) Find $\sin\frac{\theta}{2}$, $\cos\frac{\theta}{2}$, and $\tan\frac{\theta}{2}$ exactly.